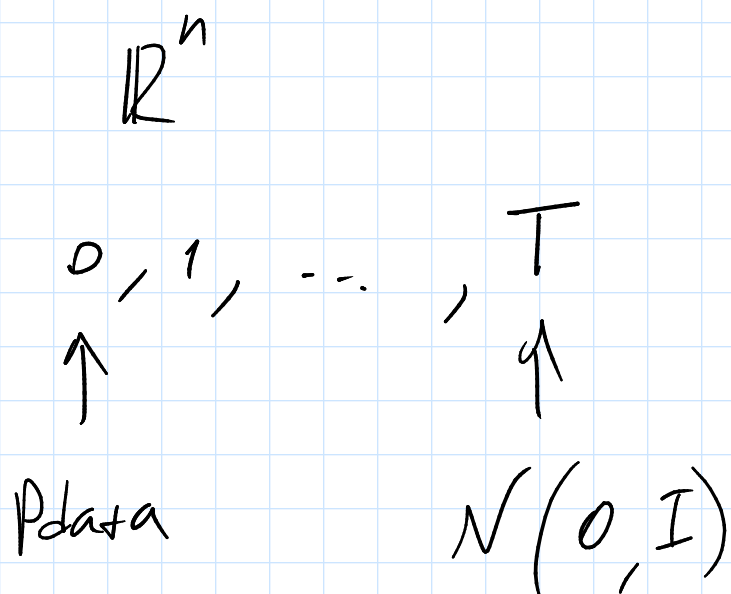


Denoising Diffusion Probabilistic Models (DDPM)

2020



Forward process

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t | \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$

transition prob. distribution

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1})$$

Backward process

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1} | \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t)$$

$\mathcal{N}(0, I)$

$\beta_t I$

1) $x_T \sim \mathcal{N}(0, I)$

2) for $t = T$ to 1:

$$x_{t-1} \sim p_{\theta}(x_{t-1} | x_t)$$

3) x_0 - final object

$$x_0 \sim p_{\text{data}}$$

$$\log p_{\theta}(x_0) = \log \int p_{\theta}(x_{0:T}) dx_1 \dots dx_T =$$

$$= \log \int q(x_{1:T} | x_0) \frac{p_{\theta}(x_{0:T})}{q(x_{1:T} | x_0)} dx_1 \dots dx_T =$$

$$= \log \mathbb{E}_{q(x_{1:T} | x_0)} \frac{p_{\theta}(x_{0:T})}{q(x_{1:T} | x_0)} \geq \left\{ \text{Jensen inequality} \right\} \geq$$

$$\geq \mathbb{E}_{q(x_{1:T} | x_0)} \log \frac{p_{\theta}(x_{0:T})}{q(x_{1:T} | x_0)} =$$



$\max_{\theta}$

$$= \mathbb{E}_{q(x_{1:T} | x_0)} \left[\log p(x_T) + \sum_{t=1}^T \log \frac{p_{\theta}(x_{t-1} | x_t)}{q(x_t | x_{t-1})} \right] = L$$

$$X_0 \rightarrow X_1 \dots X_T \quad \nabla_{\theta}$$

Training

$$1) \text{ Sample } X_1 \sim q(X_1 | X_0), \dots, X_T \sim q(X_T | X_{T-1})$$

$$2) \nabla_{\theta} \sum_{t=1}^T \log \frac{p_{\theta}(X_{t+1} | X_t)}{q(X_t | X_{t-1})}$$

$$L = \mathbb{E}_{q(X_{1:T} | X_0)} \left[-\text{KL}(q(X_T | X_1) \| p(X_T)) - \sum_{t=2}^T \text{KL}(q(X_{t-1} | X_t, X_0) \| p_{\theta}(X_{t-1} | X_t)) + \log p_{\theta}(X_0 | X_1) \right]$$

$$q(X_{t-1} | X_t, X_0) = \frac{q(X_0, X_{t-1}, X_t)}{q(X_0, X_t)} =$$

$$= \frac{\cancel{q(X_0)} q(X_{t-1} | X_0) q(X_t | X_{t-1})}{\cancel{q(X_0)} q(X_t | X_0)}$$

$$q(x_t | x_0) = \mathcal{N}\left(x_t \mid \prod_{i=1}^t \sqrt{1-\beta_i} x_0, \left(1 - \prod_{i=1}^t (1-\beta_i)\right) \mathbb{I}\right)$$

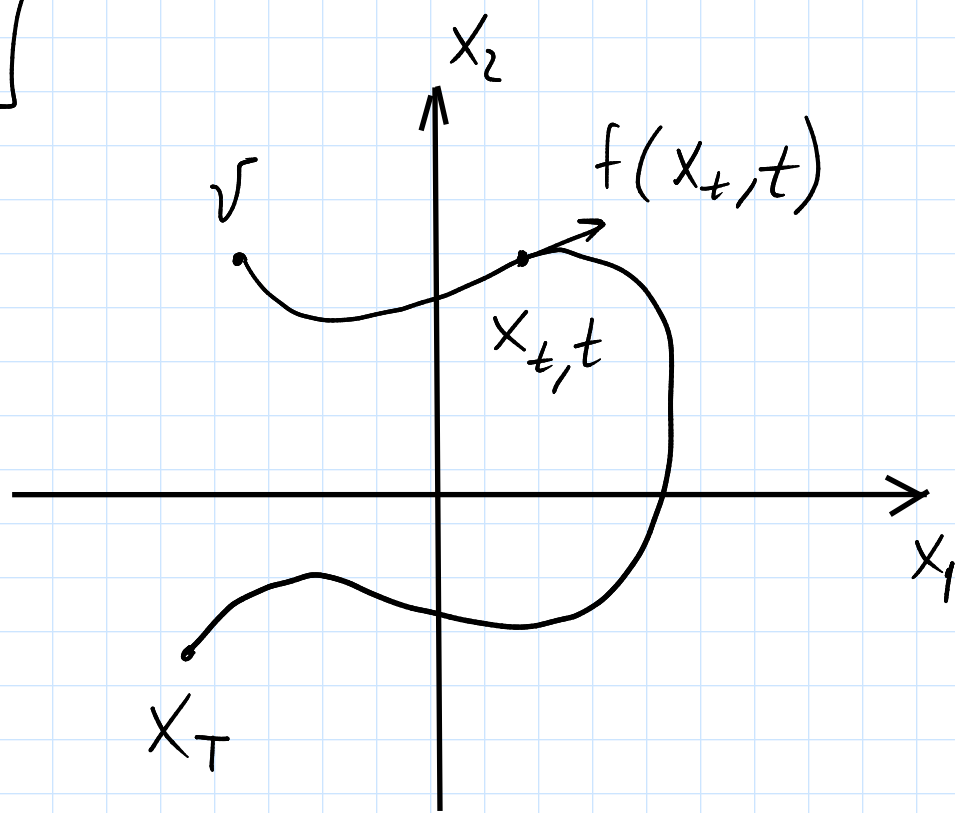
$$1) t \sim \mathcal{U}[1, \dots, T] \quad x_t \sim q(x_t | x_0)$$

$$2) \nabla_{\theta} \text{KL}(q(x_{t-1} | x_t, x_0) || p_{\theta}(x_{t-1} | x_t))$$

Ordinary Differential Equation

$$x_t \in \mathbb{R}^n \quad t \in [0, T]$$

$$\begin{cases} dx_t = f(x_t, t) dt \\ x_0 = v \in \mathbb{R}^n \end{cases}$$



Euler scheme

$$x_{t+h} = x_t + h f(x_t, t) + \bar{0}(h) \approx x_t + h f(x_t, t)$$

$$X(t)$$

Stochastic Differential Equation (SDE)

$$t \in [0, T]$$

$$\begin{cases} dX_t = f(X_t, t) dt + \underbrace{g(t)}_{\text{Standard Wiener process}} dW_t \\ X_0 \sim p_0 \end{cases}$$

Euler scheme:

$$X_{t+h} = X_t + h f(X_t, t) + \sqrt{h} g(t) \varepsilon_t \quad \varepsilon_t \sim \mathcal{N}(0, I)$$

$$X_t \sim p_t \quad \text{— solution}$$

$$t: 0 \rightarrow T$$

$$\begin{cases} dX_t = \left[f(X_t, t) - g^2(t) \nabla_x \log p_t(X_t) \right] dt + g(t) dW_t \\ X_T \sim p_T \end{cases}$$

$$t: T \rightarrow 0$$

$$\nabla_x \log p_t(X_t) \quad \text{— score function}$$

$$X_{t-h} = X_t - h f(X_t, t) + h g^2(t) \nabla_x \log p_t(X_t) + \sqrt{h} g(t) \varepsilon_t$$

SDE forward:

$$\begin{cases} dX_t = -\frac{1}{2} \beta(t) X_t dt + \sqrt{\beta(t)} dW_t \\ X_0 \sim p_{\text{data}} \end{cases}$$

$$x_+ \approx \mathcal{N}(0, I)$$

SDE backward:

$$\begin{cases} dx_t = \left(-\frac{1}{2} \beta(t) x_t dt - \beta(t) \nabla_x \log(p_t(x_t)) dt + \sqrt{\beta(t)} dW_t \right) \\ x_T \sim \mathcal{N}(0, I) \end{cases}$$

Denoising Score Matching

$$s_\theta(x, t) \approx \nabla_x \log p_t(x)$$

$$\iint p_t(x) \|s_\theta(x, t) - \nabla_x \log p_t(x)\|^2 dx dt \rightarrow \min_{\theta}$$

$\Uparrow$
 $\Downarrow$

$$\mathbb{E} \|s_\theta(x_t, t) - \nabla_x \log p_t(x_t | x_0)\|^2 \rightarrow \min_{\theta}$$

$$t \sim \mathcal{U}[0, T]$$

$$x_0 \sim p_{\text{data}}$$

$$x_t \sim p_t(x_t | x_0)$$

$h \rightarrow \text{any real value}$

DDPM

DDIM